

PYQ Questions for practice H.w
TMT CLASSES BY Shashi Kant sir

1. How many natural numbers are there between 1 and 1000 which are divisible by 5 but not by 2 (100)

2. In an A.P., the sum of first n terms is $\frac{n}{2}(3n+5)$. Find the 25th term of the A.P. (76)

3. Find the sum of all 11 terms of an A.P. whose 6th term is 30. (330)

The sum of the first three terms of an A.P. is 33. If the product of first and third term exceeds the second term by 29, find the A.P. (2, 11, 20)

4. (a) Which term of the A.P. $\frac{11}{2}, -3, \frac{1}{2}$ is $\frac{49}{2}$? (13)

(b) Find a and b so that the numbers $a, 7, b, 23$ are in A.P. (-1, 15)

5. 3. (a) In an A.P. if the sum of third and seventh term is zero, find its 5th term. (0)

OR

(b) Determine the A.P. whose third term is 5 and seventh term is 9. (3, 4, 5)

6. The n th term of the A.P. $a, 3a, 5a, \dots$ is

. (a) If the sum of first 7 terms of an A.P. is 49 and that of first 17 terms is 289, find the sum of its first 20 terms. (400)

OR

(b) The ratio of the 10th term to its 30th term of an A.P. is 1 : 3 and the sum of its first six terms is 42. Find the first term and the common difference of A.P (2)

7. If the sum of first n terms of an A.P. is $3n^2 + 4n$ and its common difference is 6, then its first term is: (7)

8. Case Study-2

Treasure Hunt is an exciting and adventurous game where participants follow a series of clues/numbers/maps to discover hidden treasures. Players engage in a thrilling quest, solving puzzles and riddles to unveil the location of the coveted prize.

While playing a treasure hunt game, some clues (numbers) are hidden in various spots collectively forming an A.P. If the number on the n th spot is $20 + 4n$, then answer the following questions to help the players in spotting the clues:

(i) Which number is on first spot? (24)

(ii) (a) Which spot is numbered as 112? (23)

Or

b) What is the sum of all the numbers on the first 10 spots?(420)

(iii) Which number is on the $(n - 2)$ th spot? $(12+4n)$

9. The ratio of the 11th term to the 18th term of an A.P. is 2: 3. Find the ratio of the 5th term to the 21st term. Also, find the ratio of the sum of first 5 terms to the sum of first 21 terms. (5:49)

If the sum of first 6 terms of an A.P. is 36 and that of the first 16 terms is 256, find the sum of first 10 terms. (100)

(10) Solve the equation:

$$-4+(-1)+2+5+ \dots + x = 437. (X = 50)$$

11.(A) Find the sum of integers between 100 and 200 which are (i) divisible by 9 (1683) (ii) not divisible by 9.(13167)

